Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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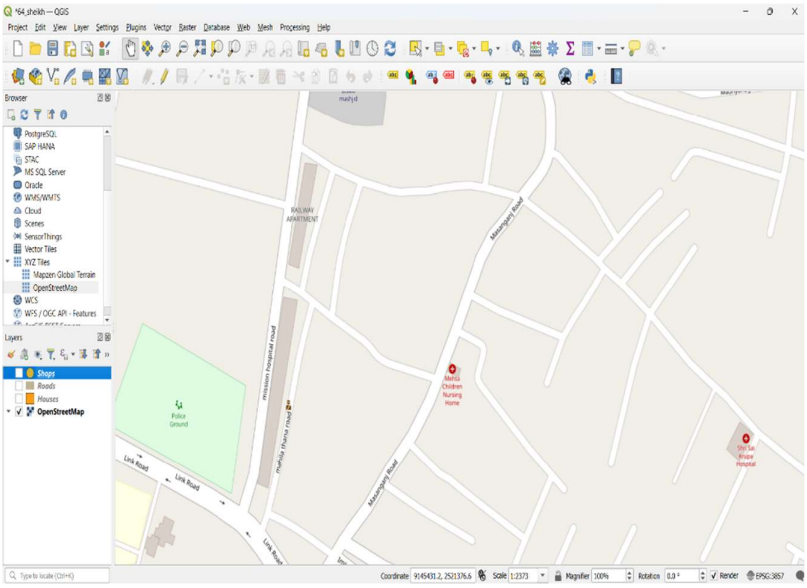
Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**03**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Create a GeoPackage or shapefiles and digitize point/line/polygon features on QGIS interface.
Problem Definition	The task requires creating a GeoPackage or shapefile in QGIS and digitizing point, line, and polygon features using the software's editing tools.
Theory (100 words)	In Geographic Information Systems (GIS), spatial data is commonly stored in structured formats such as shapefiles and GeoPackages. Both formats allow users to store and manage vector data, which represents real-world features in the form of points, lines, and polygons. 1. Data Formats o Shapefile: A traditional and widely used vector data format that stores geometry and attribute information. Although popular, shapefiles have limitations such as handling only one geometry type per file, restricted file size, and lack of support for advanced database features. o GeoPackage (GPKG): A modern, SQLite-based open standard that can store multiple layers (points, lines, polygons, and rasters) within a single file. It supports larger datasets, advanced attributes, and is more efficient compared to shapefiles. 2. Vector Feature Types o Point Features: Represent discrete locations on the earth's surface, such as schools, wells, or streetlights. o Line Features: Represent linear entities such as roads, rivers, or pipelines. o Polygon Features: Represent areas or regions such as land parcels, lakes, or city boundaries.

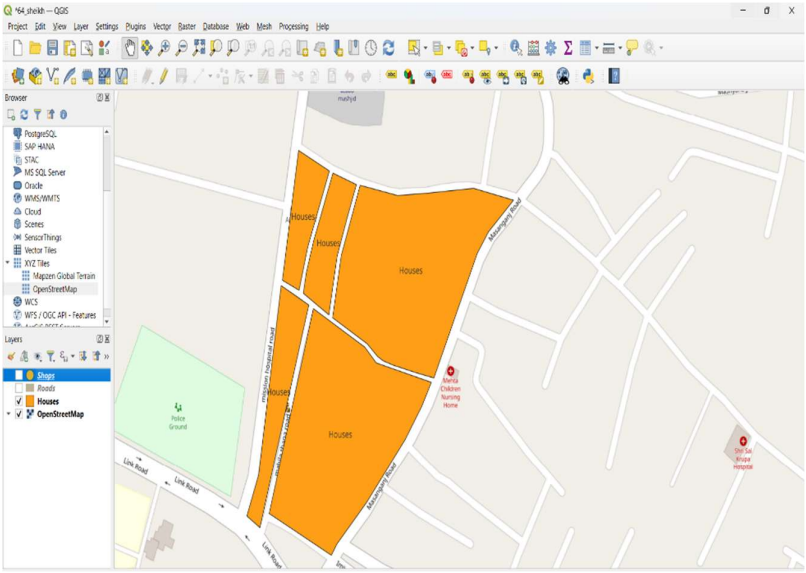
	3. Digitization in QGIS Digitization is the process of creating digital vector layers by drawing features on top of a base map or satellite imagery. In QGIS, digitization involves the following steps: o Creating a new GeoPackage or shapefile and defining its geometry type (point, line, or polygon). o Setting up the attribute table structure with relevant fields (e.g., name, ID, type). o Using the editing tools in QGIS to manually add features by clicking on the map canvas. o Saving edits to ensure that newly created features are permanently stored in the dataset. 4. Applications Digitized spatial data serves as the foundation for spatial analysis, thematic mapping, and decision-making. For instance, digitizing roads enables route planning, while polygon features of land parcels are useful in urban planning and land management.
Procedure and Execution (100 Words)	Implementation Steps: ➢ **Open QGIS** → Blank map canvas appears. ➢ **Create Layer** → Layer → Create Layer → choose **New GeoPackage** (all-in-one) or **New Shapefile** (separate files). • Select geometry type: Point / Line / Polygon. • Set CRS (use WGS 84 EPSG:4326 if unsure). • Add attribute fields (e.g., Name, ID). • Save with a clear name. ➢ **Start Editing** → Select the layer → click **pencil icon**. • Use **Add Point/Line/Polygon** tool to draw features. • Fill attribute form (e.g., Name, ID). • Save edits. ➢ **Stop Editing** → Click pencil icon again → confirm save. ➢ **Save Work** → • GeoPackage → one .gpkg file. • Shapefile → multiple files (.shp, .dbf, .shx).

Stepwise Screenshots with steps:

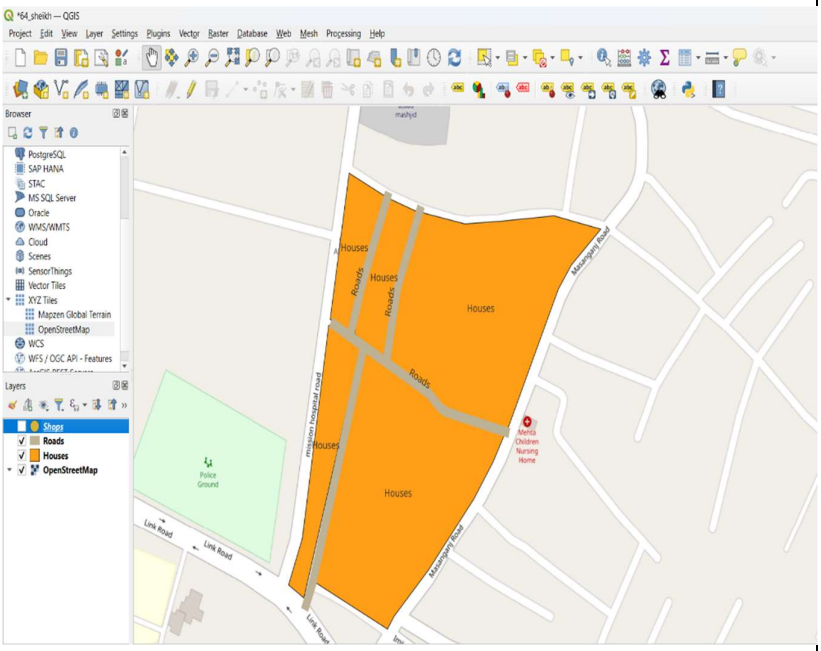
1) Open the Openstreet Map from XYZ Tiles.



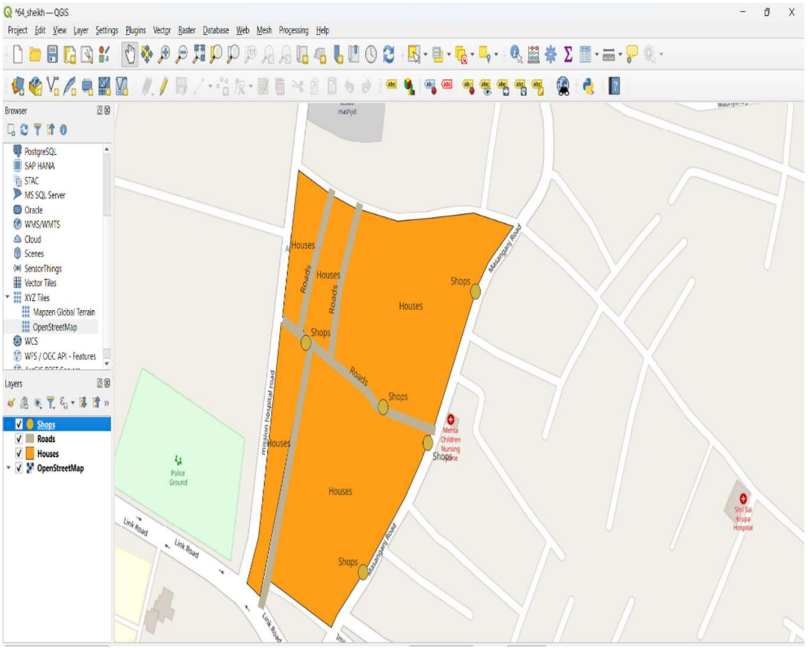
2) Now Create Layer of polygon(Houses).

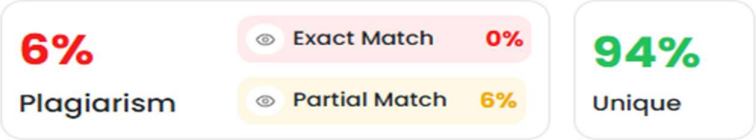


3) Similarly add for Line(Roads).



4)Now For points(shops).



Output Analysis	The practical exercise led to the creation of spatial datasets in QGIS through both GeoPackage and Shapefile formats. Different geometry types were successfully digitized: points for specific locations, lines for linear features, and polygons for area-based features. Attribute information was added to each feature, ensuring that the data carried both spatial and descriptive value. When using the GeoPackage format, all layers were stored in a single file, which provided compact and efficient data management, while the shapefile option generated multiple files for each geometry type. The digitized features were clearly displayed on the QGIS map canvas, confirming that the software can effectively support the creation, visualization, and organization of spatial information. This output demonstrates how digitization in QGIS provides a solid base for further mapping, spatial analysis, and decision-making tasks.
Conclusion	The exercise demonstrated how QGIS can be used to create and manage spatial datasets effectively. By digitizing point, line, and polygon features and adding attribute information, real-world entities were accurately represented in digital form. GeoPackage and Shapefile formats both proved useful for storing and organizing data, with GeoPackage offering a more compact and versatile solution. Overall, the task highlighted the importance of GIS tools in visualizing, analyzing, and managing spatial information for practical applications
Plag Report (Similarity index < 12%)	
Date	**12/08/2025**